# Survival Strategy Framework

## Synthetic Retention Demo

*Validation Summary (Board-Ready)*

| | |
|---|---|
| **Author** | Andrew R. Goad |
| **Document Type** | Independent Validation and Acceptance Artifact |
| **Validation Scope** | Synthetic time-to-event framework, CoxPH modeling, segmentation, scenario simulation, controls, and evidence |
| **Run ID** | 20260722T171148Z_e580c21e |
| **Run Date** | July 22, 2026 |
| **Primary Purpose** | Determine fitness for controlled portfolio demonstration and synthetic sensitivity analysis |
| **--------------------------- EXECUTIVE SNAPSHOT ---------------------------** | |
| **What Was Validated** | A deterministic Python framework spanning survival input governance, K-Means personas, regularized CoxPH modeling, risk targeting, dependency-safe same-cohort scenarios, calibration, PH review, and automated evidence. |
| **Validation Posture** | ACCEPTED WITH DOCUMENTED REVIEW for controlled synthetic strategy simulation and public portfolio demonstration. |
| **Key Evidence** | 7,500 records; 2,964 events; 5-fold CV concordance 0.827; 1,875-record target cohort; six control/improvement/stress scenarios; 42 archived artifacts. |
| **Primary Review Item** | auto_renew_flag produced a raw PH p-value of 0.0278; Holm and BH adjusted p-values were 0.333, and a stratified sensitivity model preserved scenario ranking and 93.7% of target-cohort membership. |
| **Reader Takeaway** | The system is materially more than a CoxPH model: it is a governed, reproducible, scenario-capable analytical framework with documented assumptions, controls, evidence, and explicit non-production boundaries. |
| This validation confirms fitness for the documented synthetic demonstration purpose. It does not approve production retention decisions, causal treatment claims, financial forecasts, or deployment without domain calibration and model-risk governance. | |

# 1. EXECUTIVE SUMMARY

The Survival Strategy Framework has been validated as a deterministic, configurable, and evidence-producing time-to-event decision framework. The accepted implementation separates descriptive persona discovery from Cox proportional hazards modeling, uses train-only standardization within cross-validation, selects a governed top-quartile risk cohort, applies control, favorable, and adverse scenarios to the same record identifiers, reconstructs engineered feature dependencies before re-scoring, and archives both executive and technical evidence.

The final run produced a PASS_WITH_REVIEW conclusion rather than an unrestricted pass. This is the appropriate posture: all structural, computational, reproducibility, discrimination, calibration, scenario-direction, and reporting controls passed, while one raw proportional-hazards diagnostic remained documented for review. Multiplicity-adjusted tests passed, and a stratified sensitivity analysis showed that the review item did not materially alter the decision narrative.

| Validation Dimension | Final Evidence | Conclusion |
|---|---|---|
| **Execution integrity** | Seven internal self-tests and 18 run-level acceptance checks completed; no failed checks. | Satisfied |
| **Reproducibility** | Persona assignments, coefficients, CV, calibration, PH evidence, target cohort, scenarios, and curves reproduced exactly. | Satisfied |
| **Discrimination** | 5-fold mean concordance 0.827 (SD 0.007; range 0.820-0.836). | Strong for synthetic demonstration |
| **Calibration** | Overall errors at 6, 12, and 18 months were 0.95, 1.23, and 1.39 percentage points; maximum risk-group error was 7.08%. | Satisfied |
| **PH assumptions** | One raw review item; adjusted p-values pass; stratified sensitivity preserved ranking. | Accepted with review |
| **Persona evidence** | Silhouette 0.161; mean stability ARI 0.992; descriptive segments show distinct lifecycle profiles. | Satisfied within descriptive purpose |
| **Scenario governance** | Same 1,875 IDs; no-change control neutral; improvement scenarios favorable; stress scenario adverse. | Satisfied |
| **Evidence and lineage** | 42 artifacts including configuration, model evidence, calibration, scenarios, archives, PPTX, and PDF. | Satisfied |

**FINAL VALIDATION POSTURE: Accepted with documented review for controlled synthetic time-to-event strategy simulation, educational use, and public portfolio demonstration.**

## 2. TABLE OF CONTENTS / DOCUMENT NAVIGATION

# 3. VALIDATION APPROACH OVERVIEW

## 3.1 Validation Philosophy

Validation was designed to determine whether the framework behaves as a governed analytical system, not merely whether a Python script executes. The review therefore emphasized structural integrity, reproducibility, discrimination, calibration, assumption testing, scenario attribution, engineered-feature consistency, evidence lineage, and interpretation boundaries.

- **Transparency over opacity.** Every important assumption, transformation, diagnostic, and output is archived.
- **Matched evidence over uncontrolled comparison.** Scenario movement is evaluated on the same target identifiers.
- **Out-of-sample evidence over in-sample performance alone.** Five-fold validation and out-of-fold survival predictions support discrimination and calibration review.
- **Boundary discipline over inflated claims.** Scenario effects are modeled sensitivities, not causal impacts, realized value, or production forecasts.

## 3.2 Acceptance Hierarchy

| Status | Meaning | Use |
|---|---|---|
| **PASS** | All required controls passed and no documented analytical review item remains. | Accepted within documented boundaries. |
| **PASS_WITH_REVIEW** | No failed controls; one or more diagnostics require documented interpretation or sensitivity evidence. | Accepted within documented boundaries with review item retained. |
| **FAIL** | A structural, computational, directional, reproducibility, or evidence control failed. | Not accepted until remediated and revalidated. |

## 3.3 Validation Methods

- Internal self-tests and input gatekeeping.
- Independent field-level and model-score reconciliation.
- Stratified five-fold cross-validation with fold-specific training scalers.
- Out-of-fold horizon calibration and IPCW Brier evidence.
- Raw, Holm-adjusted, and Benjamini-Hochberg PH tests.
- Scaled Schoenfeld residual review and stratified sensitivity modeling.
- Persona quality and multi-seed stability testing.
- No-change control, favorable scenarios, and adverse stress scenario.
- Exact rerun reproducibility across model, cohort, scenarios, and curves.
- Visual review of executive and technical reports.

# 4. RUN CONFIGURATION AND DATA CONTRACT

| | |
|---|---|
| **Project** | Survival Strategy Framework - Synthetic Retention Demo |
| **Engine Release** | 2026.07-mainline |
| **Run ID** | 20260722T171148Z_e580c21e |
| **Rows** | 7,500 |
| **Observed Events** | 2,964 (39.52%) |
| **Right-Censored** | 4,536 (60.48%) |
| **Duration Field** | months_observed (months) |
| **Event Field** | attrition_event |
| **Model Features** | 12 |
| **Targeting Rule** | Top 25% modeled hazard / quantile 0.75 |
| **Target Cohort** | 1,875 |
| **Input Fingerprint** | e580c21ea047d2b7d9362db41d1e1eb2fe22b5fa656adfcc59e88c46f9bb4d61 |

## 4.1 Data Contract Assessment

The dataset contains one synthetic customer-lifecycle observation per governed customer identifier. Duration is strictly positive, the event indicator is binary, model and segmentation features are numeric, and no PII or real customer records are used.

| Contract Element | Requirement | Validated Result |
|---|---|---|
| **Identifier** | Unique and non-missing | 7,500 unique IDs across 7,500 rows |
| **Duration** | Numeric, positive, bounded | Range 0.10-24.00 months |
| **Event** | Binary 0/1 | 2,964 events; 4,536 censored |
| **Model inputs** | Numeric / Boolean, finite, governed | All 12 features validated; no fit warnings were archived. |
| **Engineered terms** | Exact functions of source fields | Interactions and squared term validated before and after scenarios. |
| **Data privacy** | Synthetic / public demonstration data only | No PII or proprietary customer data. |

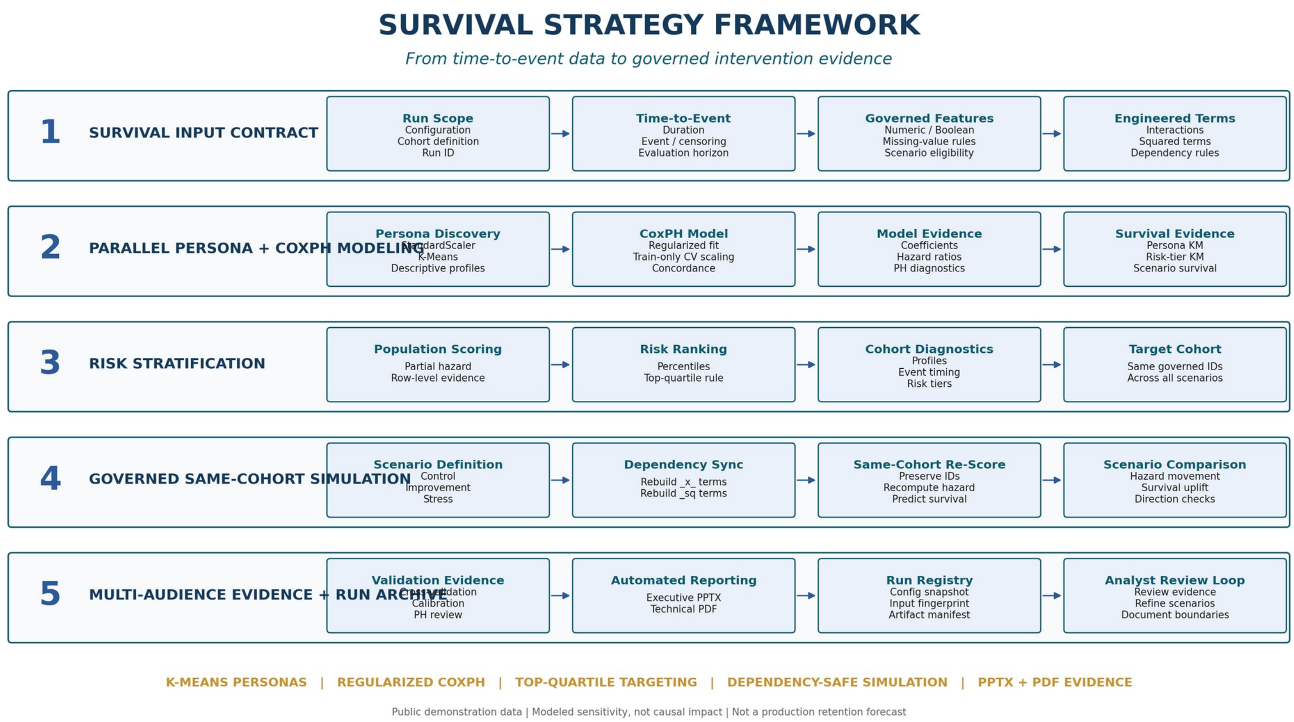


*Figure 1. Validated system architecture: input governance, parallel analytics, risk targeting, same-cohort simulation, and evidence archive.*

# 5. STRUCTURAL INTEGRITY AND REPRODUCIBILITY

## 5.1 Internal Self-Tests

| Test | Status | Evidence |
| --- | --- | --- |
| **Synthetic data reproducibility** | PASS | Same seed generated identical data. |
| **Interaction dependency synchronization** | PASS | Interaction terms rebuilt from source fields. |
| **Squared-term dependency synchronization** | PASS | Squared term rebuilt from its base field. |
| **Dependency audit status** | PASS | Post-rebuild discrepancies were zero. |
| **Nonbinary event rejection** | PASS | Invalid event value rejected before modeling. |
| **Multiplicity-adjusted PH p-values** | PASS | Adjusted values were not smaller than raw values. |
| **Scenario direction classification** | PASS | Neutral, favorable, and adverse movement classified correctly. |

## 5.2 Substantive Rerun Reproducibility

| Check | Status | Evidence |
| --- | --- | --- |
| **Persona assignments reproducible** | PASS | Segment_ID assignments were compared row-for-row under the same seed and configuration. |
| **CoxPH coefficients reproducible** | PASS | Maximum absolute coefficient difference: 0.000e+00. |
| **Cross-validation evidence reproducible** | PASS | Maximum absolute fold concordance difference: 0.000e+00. |
| **Calibration evidence reproducible** | PASS | Maximum absolute calibration difference: 0.000e+00. |
| **PH diagnostic evidence reproducible** | PASS | Maximum absolute raw/adjusted PH p-value difference: 0.000e+00; status match=True. |
| **Target cohort reproducible** | PASS | Compared 1,875 baseline target IDs with the repeated run. |
| **Scenario evidence reproducible** | PASS | Maximum absolute scenario metric difference: 0.000e+00; directions match=True. |
| **Predicted survival curves reproducible** | PASS | Maximum absolute curve difference: 0.000e+00. |
| **PH sensitivity evidence reproducible** | PASS | Maximum absolute PH sensitivity metric difference: 0.000e+00; categorical evidence match=True. |

The repeated run reproduced persona assignments, CoxPH coefficients, fold-level concordance, calibration evidence, PH raw and adjusted results, target-cohort membership, scenario metrics, survival curves, and PH sensitivity evidence with zero reported numerical difference. This exceeds a simple same-seed data check and demonstrates end-to-end analytical reproducibility.

# 6. COXPH FIT AND DISCRIMINATION VALIDATION

## 6.1 Fit Status and Model Metadata

| Metric | Observed Result | Assessment |
|---|---|---|
| **Fit status** | SUCCESS | Successful |
| **Penalizer / L1 ratio** | 0.10 / 0.00 | Regularized L2 CoxPH |
| **Apparent concordance** | 0.827 | In-sample; not used alone for validation |
| **5-fold mean concordance** | 0.827 | Out-of-sample discrimination evidence |
| **Fold SD / range** | 0.007; 0.820-0.836 | Stable across folds |
| **Partial AIC** | 46,523.7 | Archived model-fit statistic |
| **Fit warnings** | None | No captured convergence warning |

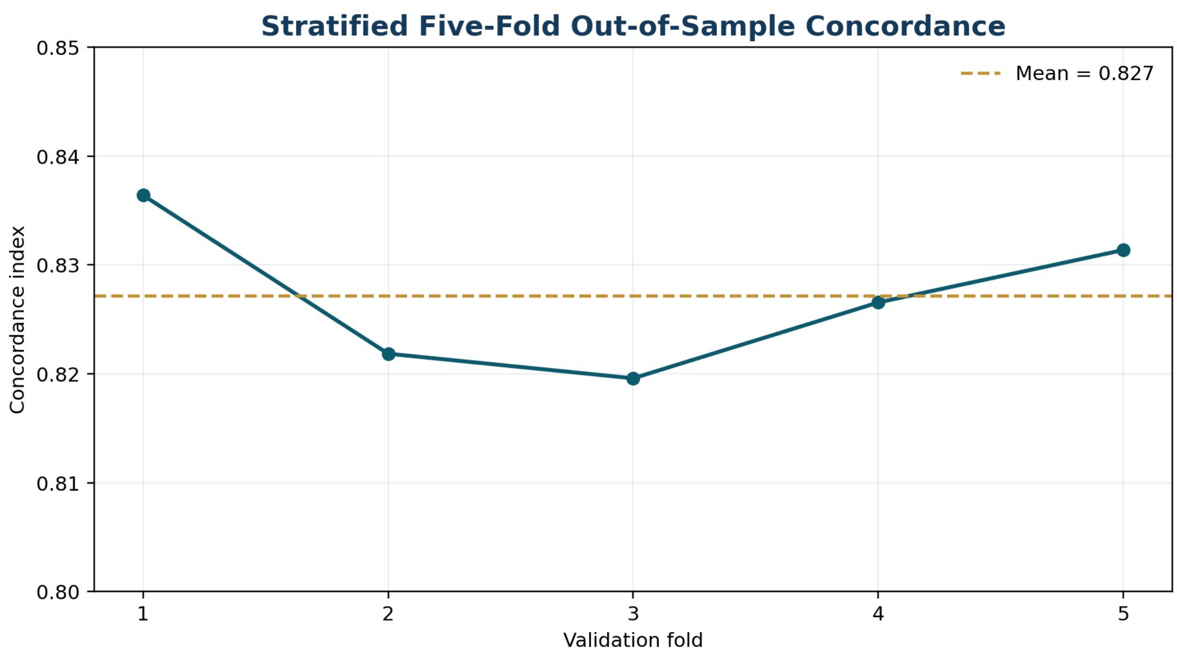


*Figure 2. Stratified five-fold validation concordance was stable across folds.*

## 6.2 Hazard Direction and Interpretability

| Feature | Coefficient | Hazard Ratio | 95% HR CI | Directional Interpretation |
|---|---|---|---|---|
| **onboarding_completion** | -0.3641 | 0.6948 | 0.6595-0.7320 | Protective |
| **monthly_active_days** | -0.2966 | 0.7433 | 0.7136-0.7743 | Protective |
| **product_adoption_count** | -0.1417 | 0.8679 | 0.8278-0.9099 | Protective |
| **support_tickets_90d** | 0.1422 | 1.1528 | 1.1042-1.2036 | Adverse |
| **support_resolution_days** | 0.2292 | 1.2575 | 1.2097-1.3073 | Adverse |
| **service_incidents_90d** | 0.1772 | 1.1938 | 1.1574-1.2314 | Adverse |
| **tenure_at_start** | -0.1109 | 0.8951 | 0.8658-0.9253 | Protective |
| **auto_renew_flag** | -0.2363 | 0.7895 | 0.7681-0.8115 | Protective |
| **enterprise_flag** | -0.1120 | 0.8940 | 0.8622-0.9270 | Protective |
| **monthly_active_days_x_product_adoption_count** | -0.2077 | 0.8125 | 0.7672-0.8604 | Protective interaction |
| **support_tickets_90d_x_support_resolution_days** | 0.2109 | 1.2348 | 1.1798-1.2923 | Adverse interaction |
| **onboarding_completion_sq** | -0.3109 | 0.7328 | 0.6931-0.7748 | Protective nonlinear term |

Because the model includes standardized linear, interaction, and squared terms, individual coefficients should be interpreted jointly with related terms. The validation conclusion concerns directional coherence and model behavior, not isolated causal effects.

## 7. CALIBRATION AND BRIER EVIDENCE

Calibration was evaluated using out-of-fold predictions, not full-sample fitted predictions. This design prevents the calibration evidence from merely restating the development fit.

| Horizon | Mean OOF Predicted | Observed KM | Difference | IPCW Brier | Interpretation |
|---|---|---|---|---|---|
| **6 months** | 74.76% | 75.71% | -0.95 pp | 0.1102 | Primary validation horizon |
| **12 months** | 63.40% | 64.63% | -1.23 pp | 0.1302 | Primary validation horizon |
| **18 months** | 56.16% | 57.55% | -1.39 pp | 0.1332 | Primary validation horizon |
| **24 months** | 49.90% | 51.25% | -1.35 pp | 0.0757 | Administrative endpoint - interpret cautiously |

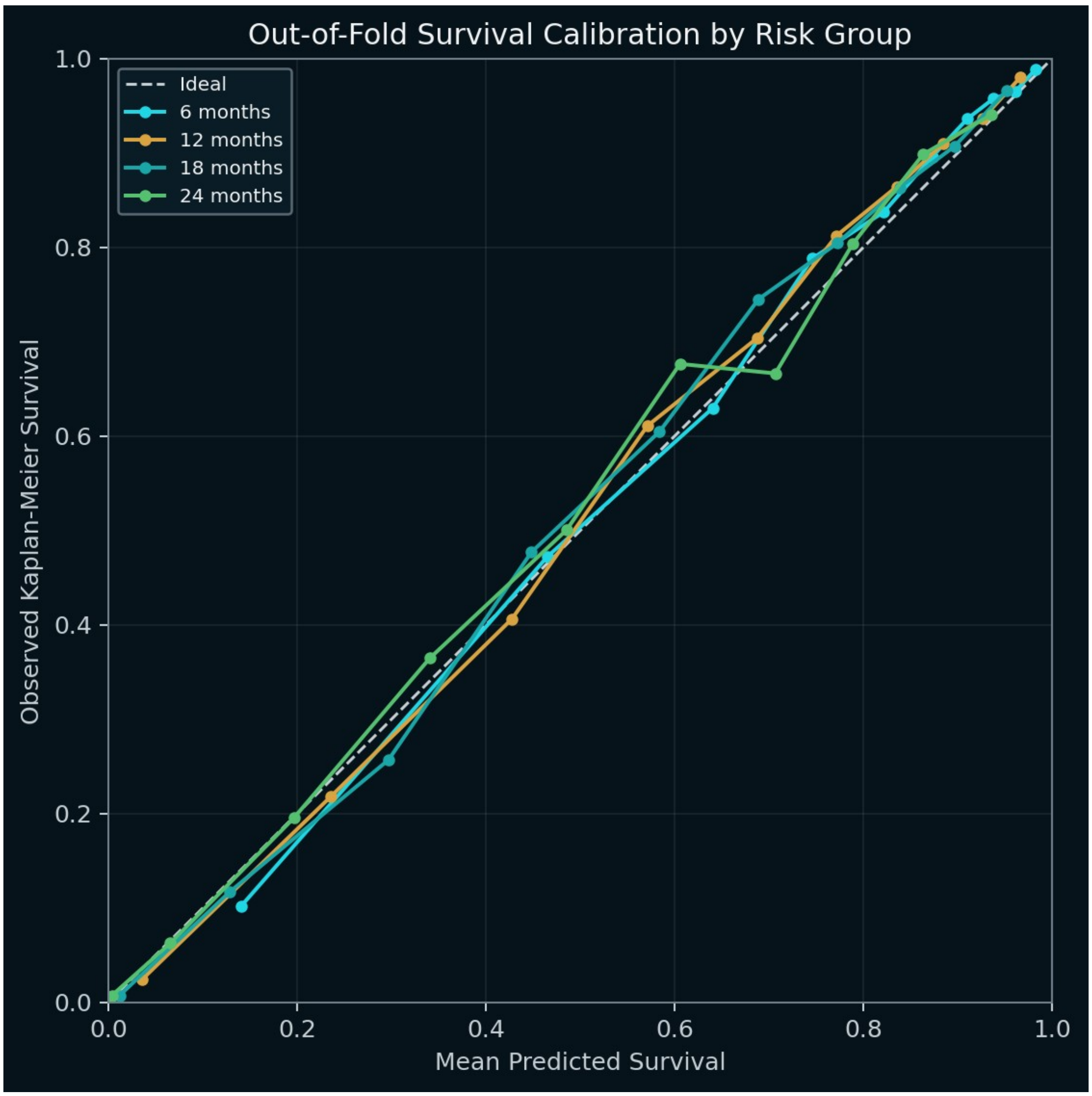


*Figure 3. Out-of-fold calibration by modeled-risk group remained close to the ideal line across configured horizons.*

### 7.1 Calibration Assessment

- Overall absolute calibration error was 0.95 pp at 6 months, 1.23 pp at 12 months, and 1.39 pp at 18 months.
- Maximum risk-group absolute calibration error was 7.08 pp, below the configured 10 pp review threshold.

- The configured-horizon Brier average was 0.119; the 6-18 month primary-horizon average was independently calculated as 0.126.
- The 24-month point coincides with the administrative follow-up boundary and has zero records remaining at risk. It is retained as endpoint evidence but is not the primary basis for calibration acceptance.

**Calibration conclusion: Accepted for synthetic demonstration. External, temporal, and domain-specific calibration remain mandatory before any production use.**

## 8. PROPORTIONAL-HAZARDS ASSUMPTION REVIEW

| Evidence | Result | Interpretation |
|---|---|---|
| **Raw PH test** | p = 0.0278 | Raw review item |
| **Holm-adjusted p-value** | 0.3331 | Pass after family-wise adjustment |
| **BH-adjusted p-value** | 0.3331 | Pass after false-discovery adjustment |
| **Sensitivity model** | Stratified CoxPH on auto_renew_flag | Appropriate low-cardinality sensitivity |
| **Sensitivity concordance** | 0.817 (delta -0.011) | Modest discrimination change |
| **Target overlap** | 93.71%; Jaccard 0.882 | High cohort stability |
| **Scenario ranking** | Spearman 1.000; exact match True | Decision narrative unchanged |

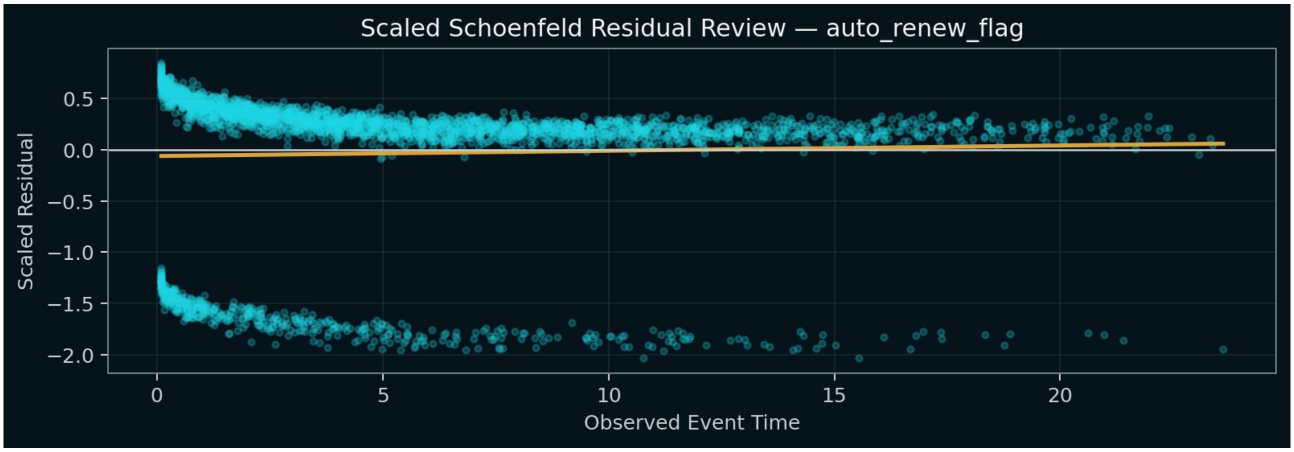


*Figure 4. Scaled Schoenfeld residual review for the raw PH review feature.*

The raw p-value is not ignored. It is retained as a documented review item, which is why the final run posture is PASS_WITH_REVIEW. However, multiplicity-adjusted tests pass, and the stratified sensitivity analysis preserves the best scenario and complete favorable-scenario ranking. The review item does not materially undermine the framework’s intended synthetic sensitivity use.

# 9. PERSONA SEGMENTATION VALIDATION

| Metric | Result | Validation Interpretation |
|---|---|---|
| **Clusters** | 3 | Three descriptive personas |
| **Silhouette score** | 0.161 | Modest separation; above configured review threshold |
| **Davies-Bouldin index** | 1.875 | Supplemental compactness/separation evidence |
| **Calinski-Harabasz index** | 1451.1 | Supplemental between/within-cluster evidence |
| **Mean stability ARI** | 0.992 | Very high multi-seed stability |
| **Minimum stability ARI** | 0.980 | No material instability |
| **Cluster size range** | 21.99%-45.45% | No trivial cluster |

| Persona | Records | Event Rate | Median Months | Onboarding | Active Days | Products | Resolution Days |
|---|---|---|---|---|---|---|---|
| **Service-Friction Risk** | 1,649 | 63.19% | 8.31 | 0.64 | 16.4 | 3.6 | 4.4 |
| **Engaged & Embedded** | 3,409 | 12.94% | 14.77 | 0.81 | 20.2 | 4.6 | 2.3 |
| **Low-Engagement Risk** | 2,442 | 60.65% | 8.92 | 0.54 | 13.6 | 2.7 | 2.6 |

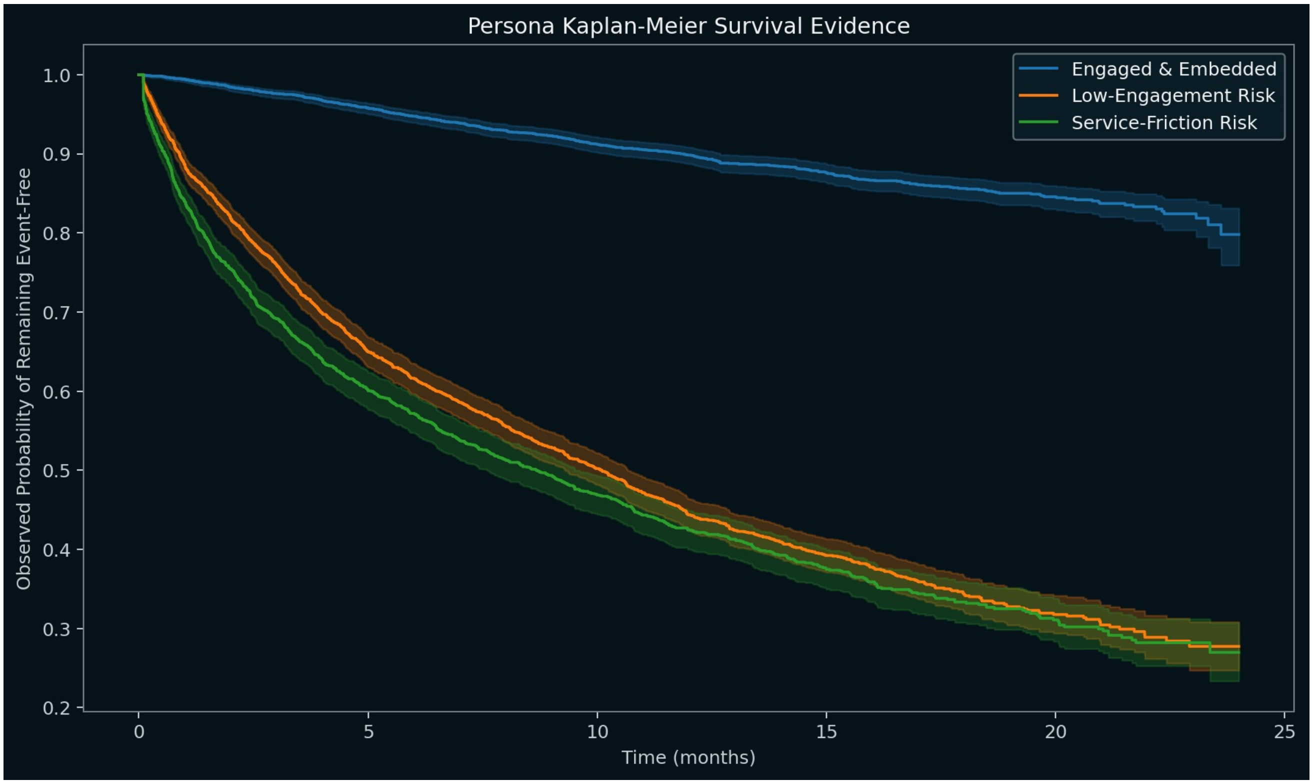


*Figure 5. Observed Kaplan-Meier survival evidence by descriptive persona.*

The silhouette score indicates that the personas are not sharply isolated natural classes; they should therefore be communicated as stable descriptive segments, not as latent truth or risk grades. Their value lies in contrasting service-friction and low-engagement pathways that produce similar event rates for different operational reasons.

# 10. RISK STRATIFICATION VALIDATION

| Risk Tier | Records | Event Rate | Median Months | Mean Hazard | Onboarding | Active Days | Resolution Days |
|---|---|---|---|---|---|---|---|
| **High Risk (Top 25%)** | 1,875 | 87.57% | 3.31 | 10.900 | 0.49 | 13.1 | 3.9 |
| **Mid Risk** | 3,750 | 31.92% | 12.71 | 1.098 | 0.69 | 17.2 | 2.8 |
| **Benchmark (Bottom 25%)** | 1,875 | 6.67% | 15.25 | 0.200 | 0.86 | 21.4 | 2.0 |

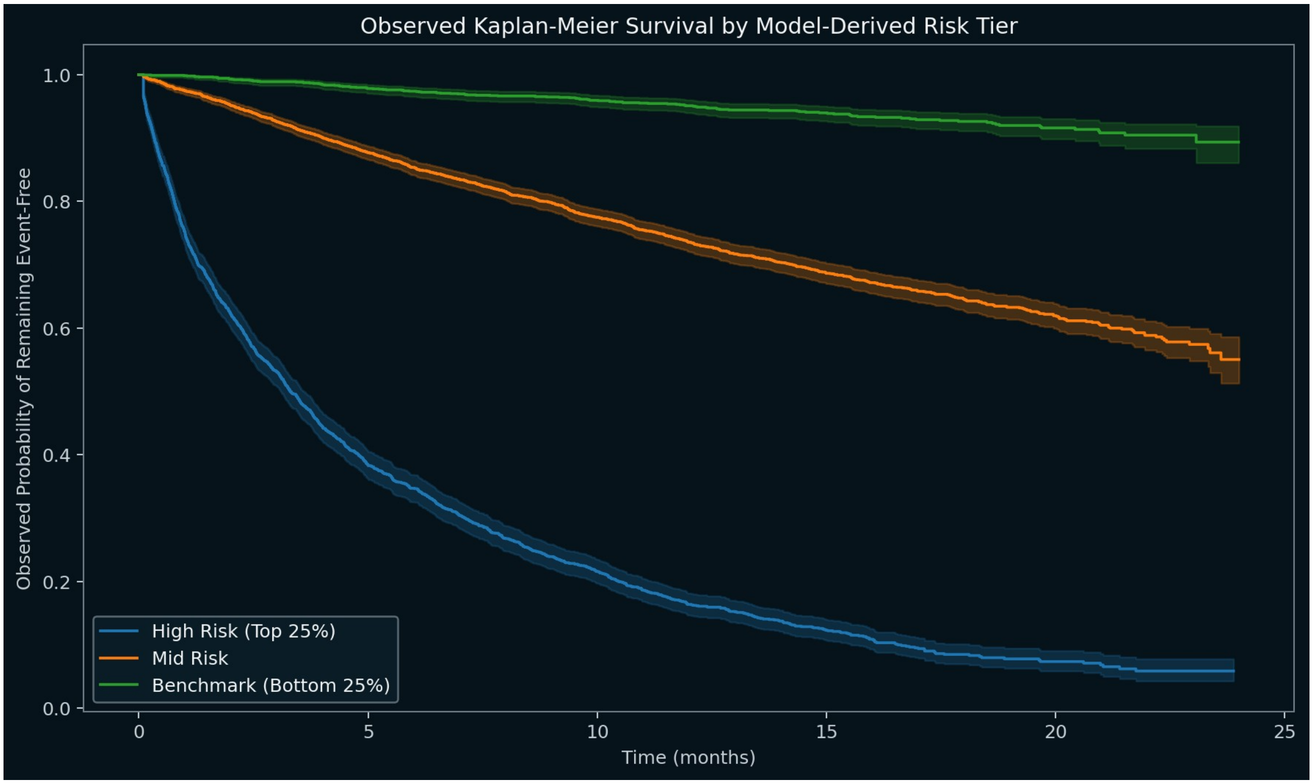


*Figure 6. Observed Kaplan-Meier survival by model-derived risk tier.*

- High-risk top quartile: event rate 87.57%, median duration 3.31 months.
- Benchmark bottom quartile: event rate 6.67%, median duration 15.25 months.
- The monotonic tier ordering is strong internal evidence, but it is derived from synthetic data and a model fitted to the full development sample. Cross-validation is reported separately to avoid overstating the tier chart as external performance evidence.

## 11. SCENARIO AND DEPENDENCY VALIDATION

| Rank | Scenario | Type | Hazard Movement | 12-Month Survival | Direction | Check |
|---|---|---|---|---|---|---|
| **1** | Combined Retention Strategy | IMPROVEMENT | +52.86% | +18.79 pp | IMPROVED | PASS |
| **2** | Onboarding Completion Improvement | IMPROVEMENT | +28.03% | +9.51 pp | IMPROVED | PASS |
| **3** | Support Resolution Improvement | IMPROVEMENT | +25.72% | +4.69 pp | IMPROVED | PASS |
| **4** | Product Adoption Expansion | IMPROVEMENT | +11.77% | +3.49 pp | IMPROVED | PASS |
| **5** | No-Change Control | CONTROL | +0.00% | +0.00 pp | NEUTRAL | PASS |
| **6** | Service Friction Stress | STRESS | -72.86% | -8.22 pp | ADVERSE | PASS |

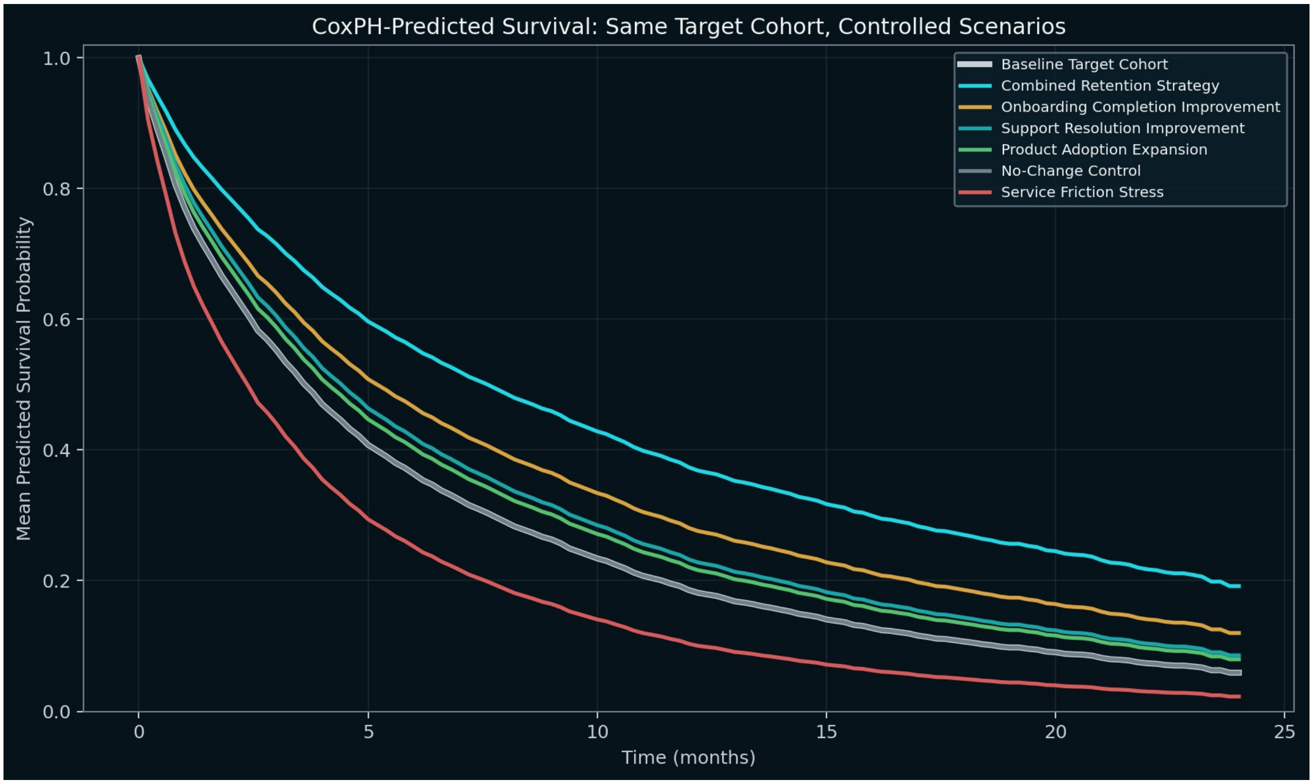


*Figure 7. CoxPH-predicted survival for the identical target cohort under control, favorable, and adverse assumptions.*

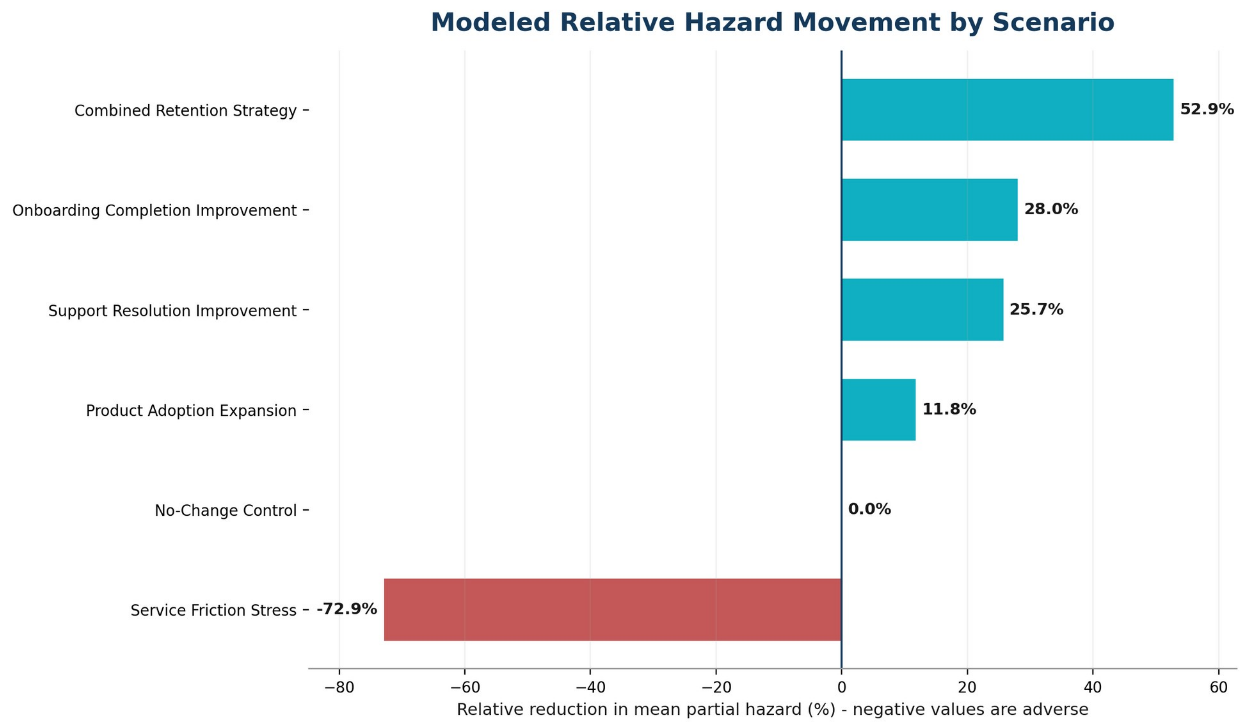


*Figure 8. Modeled relative hazard movement by scenario; positive values are favorable and negative values are adverse.*

## 11.1 Control and Direction Testing

- No-Change Control produced exactly neutral hazard and survival movement within numerical tolerance.
- Every improvement scenario produced favorable movement in both mean partial hazard and 12-month predicted survival.
- Service Friction Stress produced adverse movement: mean partial hazard increased 72.86% relative to baseline, and 12-month modeled survival declined 8.22 percentage points.
- Scenario ranking and direction checks passed for all six scenarios.

## 11.2 Scenario Change and Bound Audit

| Scenario | Feature | Operation | Config | Changed | Lower Bound | Upper Bound | Mean Change |
|---|---|---|---|---|---|---|---|
| **No-Change Control** | NO_CHANGE | NONE | 0.0 | 0.00% | 0.00% | 0.00% | 0.0000 |
| **Onboarding Completion Improvement** | onboarding_completion | add | 0.1 | 100.00% | 0.00% | 0.16% | 0.1000 |
| **Support Resolution Improvement** | support_resolution_days | multiply | 0.75 | 99.63% | 0.43% | 0.00% | -0.9692 |
| **Product Adoption Expansion** | product_adoption_count | add | 1.0 | 99.52% | 0.00% | 2.13% | 0.9952 |
| **Combined Retention Strategy** | onboarding_completion | add | 0.1 | 100.00% | 0.00% | 0.16% | 0.1000 |
| **Combined Retention Strategy** | support_resolution_days | multiply | 0.75 | 99.63% | 0.43% | 0.00% | -0.9692 |
| **Combined Retention Strategy** | product_adoption_count | add | 1.0 | 99.52% | 0.00% | 2.13% | 0.9952 |
| **Service Friction Stress** | support_resolution_days | multiply | 1.25 | 100.00% | 0.00% | 0.00% | 0.9695 |
| **Service Friction Stress** | service_incidents_90d | add | 1.0 | 100.00% | 0.00% | 0.00% | 1.0000 |

### 11.3 Dependency Synchronization

All 18 dependency-audit rows passed. The maximum pre-rebuild discrepancies were 25.3890; every post-rebuild discrepancy was 0.0000. This confirms that scenario changes did not leave stale interaction or squared terms in the model matrix.

## 12. OUTPUT AND ARTIFACT VALIDATION

The run registry contains 42 named artifacts. The archive preserves the configuration, input fingerprint, input snapshot, validation checks, model fit, feature scaler, coefficients, PH diagnostics, cross-validation, calibration, personas, risk tiers, target cohort, scenario definitions, row-level scenario scores, dependency audit, predicted curves, executive narrative, PowerPoint deck, technical PDF, and artifact manifest.

| Artifact Family | Contents |
|---|---|
| **Governance** | framework_config.json; run_metadata.json; scenario_definitions.json; artifact_manifest.json |
| **Input evidence** | input_snapshot.csv; input_validation.csv; input fingerprint |
| **Model evidence** | model_summary.csv; model_fit_metadata.csv; scaler parameters; baseline survival and cumulative hazard |
| **Validation evidence** | cross-validation; calibration; PH tests; residuals; sensitivity; persona quality; reproducibility |
| **Scenario evidence** | target cohort; scenario results; scenario scores; scenario change audit; dependency audit; predicted curves |
| **Stakeholder outputs** | Survival_Strategy_Deck.pptx; Technical_Model_Evidence.pdf; executive narrative |

### 12.1 Visual and Reporting Review

- The executive deck contains eight slides and no clipped scenario table or chart after the scenario table and hazard movement were separated.
- The technical evidence PDF contains 14 pages of validation evidence, including raw and adjusted PH tests, sensitivity review, persona quality, scenario audit, dependency reconstruction, and reproducibility.
- A minor presentation limitation remains in the code-generated adverse bar label at full chart size; the underlying value and table are correct. This is cosmetic and does not affect model acceptance.

# 13. SYSTEM-LEVEL VALIDATION SYNTHESIS

| Validation Dimension | Evidence | Assessment |
|---|---|---|
| **Governed input contract** | Unique IDs, positive duration, binary events, numeric features, explicit configuration, and fail-fast validation. | Satisfied |
| **Independent analytical branches** | K-Means personas remain separate from the CoxPH model feature matrix. | Satisfied |
| **Model performance** | Stable 5-fold concordance and out-of-fold calibration. | Satisfied |
| **Assumption governance** | Raw PH review retained; adjusted tests and sensitivity analysis support stability. | Accepted with review |
| **Targeting** | Top-quartile cohort contains exactly 1,875 governed IDs. | Satisfied |
| **Scenario attribution** | Same target IDs across control, improvement, and stress scenarios. | Satisfied |
| **Mathematical consistency** | Engineered interactions and squared terms rebuilt with zero post-rebuild discrepancy. | Satisfied |
| **Bidirectional sensitivity** | No-change is neutral, favorable scenarios improve, stress scenario worsens. | Satisfied |
| **Evidence lineage** | 42-artifact run registry supports reconstruction and review. | Satisfied |
| **Boundary discipline** | Synthetic, non-causal, non-production limitations are repeated in code and reports. | Satisfied |

**System-level conclusion: the framework is fit for its stated purpose as a governed, reproducible, scenario-capable survival-analysis demonstration and teaching artifact.**

# 14. MODEL BOUNDARIES AND LIMITATIONS

| Supported Use | Validation Support |
|---|---|
| **Synthetic time-to-event data demonstration** | Input contract, censoring, event design, and deterministic generator validated. |
| **Descriptive lifecycle personas** | Stable K-Means segments with profile and Kaplan-Meier evidence. |
| **Relative-risk modeling** | Regularized CoxPH, cross-validation, hazard ratios, and PH diagnostics. |
| **Controlled sensitivity analysis** | Same-cohort control, improvement, and stress scenarios. |
| **Executive and technical evidence** | Automated PPTX, PDF, run metadata, and CSV/JSON archive. |
| **Teaching and portfolio demonstration** | Detailed code comments, explicit assumptions, and model boundaries. |

| Explicit Non-Use | Reason |
|---|---|
| **Production retention decisions** | No real customer data, approved strategy, implementation testing, or operating governance. |
| **Causal intervention claims** | Scenarios modify model inputs; they do not identify treatment effects. |
| **Realized revenue / LTV / ROI** | No observed financial outcomes or validated value model. |
| **External population prediction** | No temporal, external, or institution-specific validation. |
| **Regulatory model validation substitute** | Portfolio validation evidence does not replace formal independent model-risk review. |
| **Automated customer treatment** | Outputs are analytical evidence, not operational instructions. |

## 14.1 Endpoint Calibration Limitation

The 24-month calibration point coincides with the synthetic administrative follow-up boundary, where zero records remain at risk. The point is useful as endpoint context but should not be treated as the primary basis for Brier or calibration acceptance. The 6-, 12-, and 18-month out-of-fold evidence provides the primary calibration conclusion. A future enhancement should automatically flag or exclude horizons without adequate at-risk support.

## 15. FUTURE ENHANCEMENT ROADMAP

| Priority | Enhancement | Rationale |
|---|---|---|
| **Near-term** | At-risk support gate for calibration horizons | Automatically flag or omit evaluation horizons with insufficient censoring support. |
| **Near-term** | Cosmetic adverse-label placement | Improve chart label positioning for negative values in generated reports. |
| **Analytical** | Temporal or external validation | Evaluate transportability beyond the deterministic synthetic population. |
| **Analytical** | Time-varying effects | Model features whose effects change materially over time. |
| **Analytical** | Competing risks | Support mutually exclusive event types where relevant. |
| **Governance** | Scenario cost / feasibility layer | Add operational cost and feasibility before describing a scenario frontier. |
| **Governance** | Model monitoring package | Add drift, stability, recalibration, and ongoing performance controls. |
| **Production boundary** | Causal intervention design | Requires experimental or quasi-experimental identification, separate from sensitivity modeling. |

These items are additive enhancements. They do not invalidate the current acceptance for controlled synthetic demonstration use.

# 16. FINAL VALIDATION STATEMENT

The Survival Strategy Framework has achieved its validation objective. It produces a stable, deterministic synthetic time-to-event population; separates descriptive segmentation from CoxPH risk modeling; generates out-of-fold discrimination and calibration evidence; selects a governed high-risk cohort; applies neutral, favorable, and adverse scenarios to identical record identifiers; reconstructs dependent engineered terms; and archives decision-ready and technical evidence.

The raw proportional-hazards review item for auto_renew_flag is documented rather than suppressed. Multiplicity-adjusted tests pass, and the stratified sensitivity analysis preserves the favorable-scenario ranking and best-scenario conclusion. Accordingly, the appropriate final posture is accepted with documented review, not an unrestricted production approval.

**FINAL VALIDATION POSTURE: ACCEPTED WITH DOCUMENTED REVIEW for controlled synthetic survival strategy simulation, public portfolio demonstration, and educational use within the stated boundaries.**

The framework is not accepted or represented as a production retention model, causal intervention engine, realized financial-value model, regulatory validation substitute, or customer-treatment system.

# APPENDIX A. RUN-LEVEL ACCEPTANCE CHECKLIST

| Check | Status | Evidence |
| --- | --- | --- |
| **Unique governed IDs** | PASS | 7,500 unique IDs across 7,500 rows. |
| **Target cohort is non-empty** | PASS | 1,875 records selected at quantile 0.75. |
| **Same target cohort across scenarios** | PASS | Every scenario score table was compared with the identical baseline target IDs. |
| **Engineered dependencies synchronized** | PASS | All interaction and squared-term post-rebuild checks were reviewed. |
| **Scenario change audit completed** | PASS | Configured changes, realized movement, and bound-hit rates were archived. |
| **Scenario metrics are calculated and finite** | PASS | No hard-coded uncertainty, ROI, revenue, or defended-value metric is used. |
| **Predicted survival remains within [0, 1]** | PASS | All baseline and scenario survival probabilities satisfy probability bounds. |
| **Predicted survival begins at S(0)=1** | PASS | The first baseline and scenario survival values were tested at time zero. |
| **Scenario direction matches configured expectation** | PASS | Control, favorable, stress, and technical scenarios were compared with expected direction. |
| **No-change control scenario** | PASS | Expected zero modeled movement within numerical tolerance. |
| **Adverse stress scenario** | PASS | Expected negative hazard-reduction and survival-uplift evidence. |
| **Cross-validation completed** | PASS | 5 stratified folds were required. |
| **Out-of-fold calibration evidence** | PASS | Maximum group absolute calibration error: 7.08%. |
| **Persona quality evidence** | PASS | Silhouette=0.161; mean stability ARI=0.992. |
| **Proportional-hazards diagnostics** | REVIEW | Global documented PH status: PASS_WITH_REVIEW. |
| **PH sensitivity review** | PASS | Low-cardinality raw PH review features were stratified where technically appropriate; target-cohort overlap and favorable-scenario ranking were compared. |
| **Substantive reproducibility** | PASS | Persona, model, validation, target-cohort, scenario, and curve outputs were repeated. |
| **Executive and technical reports created** | PASS | Survival_Strategy_Deck.pptx \| Technical_Model_Evidence.pdf |

# APPENDIX B. INDEPENDENT SCORE RECONCILIATION

Independent reconciliation was performed outside the framework by rebuilding standardized model matrices from the archived input snapshot, scaler parameters, and CoxPH coefficients, then recomputing partial hazards for the target cohort and all six scenarios.

| Reconciliation Test | Maximum Absolute Difference | Conclusion |
|---|---|---|
| **Baseline target-cohort partial hazard** | 5.684e-13 | Machine-precision agreement |
| **Scenario partial hazards** | 2.728e-12 | Machine-precision agreement |
| **Scenario mean partial hazard and hazard reduction** | < 1e-12 | Exact within floating-point tolerance |
| **No-change control** | 0 modeled movement | Neutral control validated |
| **Dependency post-rebuild discrepancy** | 0.0000 for every row | Engineered terms synchronized |

This independent reconstruction confirms that archived scenario results are mathematically consistent with the fitted coefficients, saved scaler, governed feature changes, and dependency rules.

# APPENDIX C. ARTIFACT INVENTORY

| Category | Artifact |
| --- | --- |
| **Stakeholder Output** | Survival_Strategy_Deck.pptx |
| **Stakeholder Output** | Technical_Model_Evidence.pdf |
| **Run Evidence** | acceptance_checks.csv |
| **Run Evidence** | calibration_by_risk_group.csv |
| **Run Evidence** | calibration_metrics.csv |
| **Chart** | charts/baseline_vs_scenario_survival.png |
| **Chart** | charts/calibration_at_horizons.png |
| **Chart** | charts/persona_kaplan_meier.png |
| **Chart** | charts/ph_assumption_diagnostics.png |
| **Chart** | charts/risk_tier_kaplan_meier.png |
| **Chart** | charts/scenario_hazard_reduction.png |
| **Run Evidence** | cox_baseline_cumulative_hazard.csv |
| **Run Evidence** | cox_baseline_survival.csv |
| **Run Evidence** | cross_validated_predictions.csv |
| **Run Evidence** | cross_validation_results.csv |
| **Run Evidence** | dependency_audit.csv |
| **Run Evidence** | executive_narrative.txt |
| **Run Evidence** | framework_config.json |
| **Run Evidence** | input_snapshot.csv |
| **Run Evidence** | input_validation.csv |
| **Run Evidence** | model_fit_metadata.csv |
| **Run Evidence** | model_fit_warnings.csv |
| **Run Evidence** | model_scaler_parameters.csv |
| **Run Evidence** | model_summary.csv |
| **Run Evidence** | persona_centroids_raw_scale.csv |
| **Run Evidence** | persona_centroids_standardized.csv |
| **Run Evidence** | persona_profiles.csv |
| **Run Evidence** | persona_quality_metrics.csv |
| **Run Evidence** | persona_scaler_parameters.csv |
| **Run Evidence** | persona_stability_results.csv |
| **Run Evidence** | ph_sensitivity_results.csv |
| **Run Evidence** | predicted_survival_curves.csv |

| Category | Artifact |
| --- | --- |
| **Run Evidence** | proportional_hazards_test.csv |
| **Run Evidence** | reproducibility_checks.csv |
| **Run Evidence** | risk_tier_profiles.csv |
| **Run Evidence** | run_metadata.json |
| **Run Evidence** | scaled_schoenfeld_residuals.csv |
| **Run Evidence** | scenario_change_audit.csv |
| **Run Evidence** | scenario_definitions.json |
| **Run Evidence** | scenario_results.csv |
| **Run Evidence** | scenario_target_scores.csv |
| **Run Evidence** | target_cohort.csv |